LAB EXPERIMENT RECORD

15-10-2026---Set 21: Energy Consumption Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Day	THURSDAY

Aim

To analyze energy consumption data in R using a histogram with density plot, a bubble scatter plot and a bar chart, and to build an interactive Tableau dashboard.

Dataset: Energy_Consumption_Data.csv

Sector	Region	Month	Temp. (°C)	Units Consumed (kWh)	Cost (₹)	Renewable Usage (%)	Peak Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

Question 1: Histogram and Density Plot of Units Consumed

Question: Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

Aim: To study how energy consumption is distributed and to compare the consumption of the sectors using a histogram and a density plot.

```r
data <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
             "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

print(data)
```

```
+   Region = c("North", "South", "West", "East", "North", "South"),
+   Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
+   Temperature = c(15, 24, 20, 18, 28, 30),
+   Units_Consumed = c(320, 540, 880, 350, 610, 920),
+   Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
+   Renewable_Usage = c(22, 18, 12, 25, 20, 15),
+   Peak_Hours = c(4, 6, 8, 5, 7, 9)
+ )
>
> print(data)
       Sector Region Month Temperature Units_Consumed Cost Renewable_Usage Peak_Hours
1 Residential  North   Jan          15            320 2100              22          4
2  Commercial  South   Jan          24            540 3600              18          6
3  Industrial   West   Feb          20            880 5900              12          8
4 Residential   East   Feb          18            350 2300              25          5
5  Commercial  North   Mar          28            610 4100              20          7
6  Industrial  South   Mar          30            920 6200              15          9
> )
```

Result: Units range from 320 to 920 with a mean of about 603, and the values form three groups: Residential (about 335), Commercial (about 575) and Industrial (about 900), so the density plot shows more than one peak.

Question 2: Bubble Scatter Plot of Temperature vs Units Consumed

Question: Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

Aim: To explore how temperature and peak hours relate to energy use using a transparent bubble scatter plot.

```r
library(ggplot2)

data <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
             "Residential", "Commercial", "Industrial"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

print(data)
```

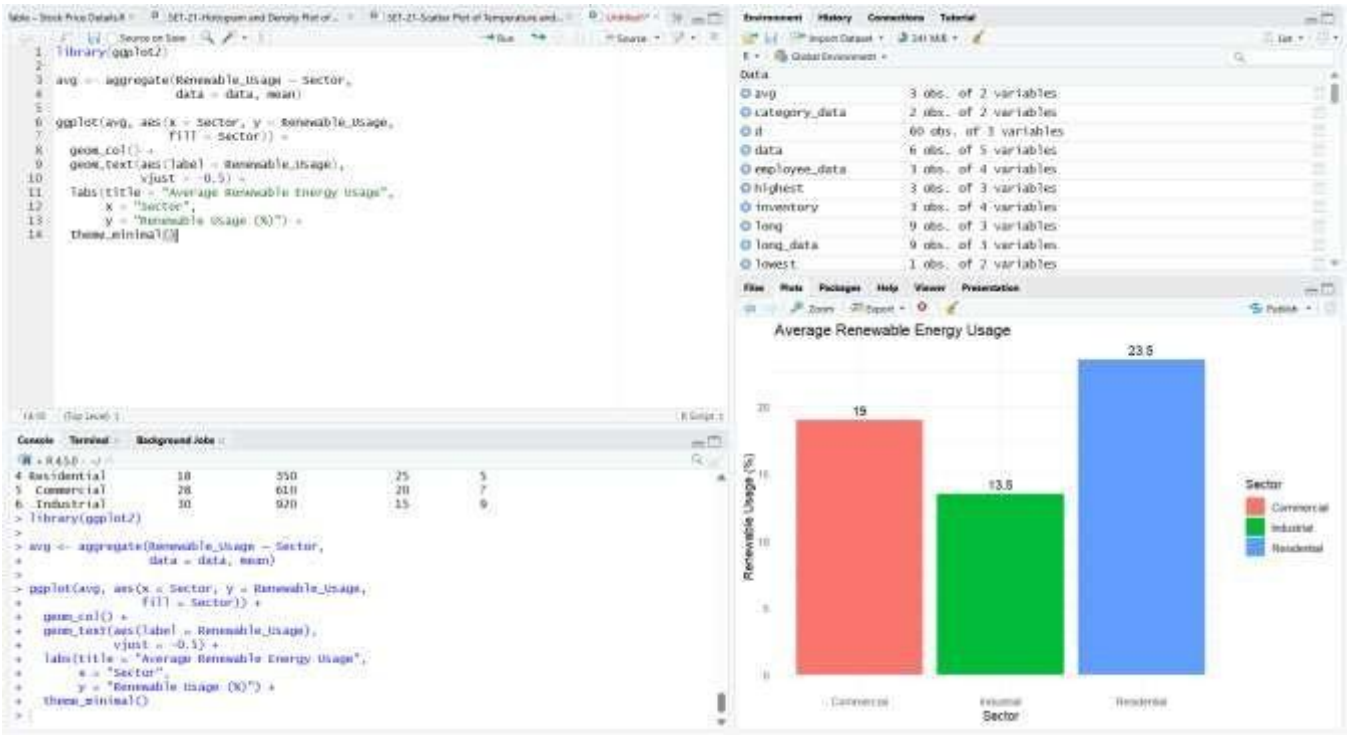
```
> data <- data.frame(
+   Sector = c("Residential", "Commercial", "Industrial",
+              "Residential", "Commercial", "Industrial"),
+   Temperature = c(15, 24, 20, 18, 28, 30),
+   Units_Consumed = c(320, 540, 880, 350, 610, 920),
+   Renewable_Usage = c(22, 18, 12, 25, 20, 15),
+   Peak_Hours = c(4, 6, 8, 5, 7, 9)
+ )
>
> print(data)
       Sector Temperature Units_Consumed Renewable_Usage Peak_Hours
1 Residential          15            320              22          4
2  Commercial          24            540              18          6
3  Industrial          20            880              12          8
4 Residential          18            350              25          5
5  Commercial          28            610              20          7
6  Industrial          30            920              15          9
>
```

Result: Units rise with temperature (correlation about 0.65), and the bubbles with more peak hours are the high-consumption points (correlation about 0.98), so peak hours explain consumption best.

Question 3: Average Renewable Usage by Sector

Question: Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

Aim: To calculate the average renewable usage of each sector and show it in a bar chart.



Result: Residential averages 23.5%, Commercial 19.0% and Industrial 13.5%, so renewable adoption is highest in homes and lowest in industry, which uses the most energy.

Question 4: Interactive Tableau Dashboard

Question: Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

Aim: To build an interactive Tableau dashboard with a monthly usage line chart, a sector-region heatmap and filters for Month and Sector.



Website Traffic Details

Date	Home Page Views	Products Page Views	Blog Page Views	Contact Page Views	Total Page Views
01-Jan-2023	1,200	800	500	300	2,800
02-Jan-2023	1,450	950	650	350	3,400
03-Jan-2023	1,300	900	600	320	3,120
04-Jan-2023	1,600	1,000	700	400	3,700
05-Jan-2023	1,700	1,150	750	450	4,050
Total	**7,250**	**4,800**	**3,200**	**1,820**	**17,070**

Result: Monthly usage rises from 860 in January to 1230 in February and 1530 in March, and the heatmap's darkest cells are Industrial in South (920) and West (880); the filters update both views.

Overall Result

Observation: Total consumption is 3620 kWh, the average temperature is 22.5 °C and the average renewable usage is about 18.7%, with Industrial the largest user and the smallest adopter of renewables.